

Room Name: The Greenholt Phish

Date: August 2026

Category: Phishing

Tools Used: WHOIS, DMARCIAN SPF Lookup, MXToolbox (DMARC), VirusTotal

Scenario

In this scenario or simulation, we are a SOC Analyst that have received a suspicious email that a sales executive has sent to us. Our task is to investigate and find further evidence of the email being malicious using various tools. The SPF, DKIM, and DMARC of an email will be investigated to determine signs of malware.

Investigation

Once the virtual environment was set up, I opened the suspicious email called challenge.eml.

webmaster@redacted.org your: Transfer Reference Number:(09674321...

File Edit View Go Message Tools Help

Get Messages Write Tag

Reply Reply All Forward Archive Junk Delete More

Mr. James Jackson
info@mutawamarine.com

To webmaster@redacted.org

6/10/20, 05:58

Reply to Mr. James Jackson <info.mutawamarine@mail.com>

webmaster@redacted.org your: Transfer Reference
Number:(09674321)

Good day webmaster@redacted.org ,

**As instructed, funds has been transferred to your account this morning
via SWIFT.**

Details are as below and a receipt of payment is attached.

Interbank Transfer Reference Number: 09674321

Transaction Status: Successful

Transaction Date / Time: 10-06-2020 09:18:55

Transaction Description: Balance / Final Payment

From Account: 3105234819

Amount: 149,650

Currency: usd

> 1 attachment: SWT_#09674321__PDF__.CAB 400 KB

Save

Upon inspection, I saw a few details. The Transfer Reference Number was 09674321. The sender was Mr. James Jackson with the email address info@mutawamarine.com. There is also an attachment in the email, which will be looked at towards the end of the investigation. Finding information on the email is important as the first step in an investigation. When inspecting this email in detail through its source (CTRL + U), by looking at the earliest/oldest Receiving header near the bottom of the email headers, it was clear that the originating IP address of this email was 192.119.71.157.

```
Received: from hwsrv-737338.hostwindsdns.com ([192.119.71.157]:51810 helo=mutaw  
by sub.redacted.com with esmtp (Exim 4.80)  
(envelope-from <info@mutawamarine.com>)  
id 1jissD-0004g5-Ts  
for webmaster@redacted.org; Wed, 10 Jun 2020 01:03:04 -0400
```

To attribute the discovered IP address to its owner, I opened the *WHOIS* tool. The *WHOIS* tool revealed the owner of the IP address to be HostPapa.

```
Organization: HostPapa (HOSTP-7)  
RegDate: 2012-11-12  
Updated: 2026-05-13  
Comment: Geofeed https://geofeeds.oniaas.io/geofeeds.csv  
Ref: https://rdap.arin.net/registry/ip/192.119.64.0  
OrgName: HostPapa
```

A Sender Policy Framework (SPF) is a framework used to authenticate the sender of an email and is a key indicator that lets one know if an email is legitimate or not. Before checking the SPF of the email, its Return-Path domain has to be identified. When investigating the email's source, the Return-Path domain is discovered to be [mutawamarine.com](mailto:info@mutawamarine.com).

```
Return-Path: <info@mutawamarine.com>
```

After discovering the email's Return-Path domain, I opened up dmarcian.com, an SPF lookup tool. The tool revealed the domain's SPF record to be valid.

High five! Your SPF record is valid, but...

I also got a full SPF record of `v=spf1 include:spf.protection.outlook.com -all`. Breaking down this full record, `v=spf1` is the start of the SPF record, while the `include` tag indicates how outlook is the only legitimate domain allowed to send out emails, with the `-all` tag indicating that any other non-authorized email will be rejected.

For a DMARC lookup, I used the MXToolbox tool. DMARC ensures the sender’s domain matches the domains verified by both SPF and DKIM. After entering the discovered Return-Path domain, MXToolbox identified the full DMARC lookup as `v=DMARC1; p=quarantine; fo=1`.

`v=DMARC1; p=quarantine; fo=1`

Tag	TagValue	Name	Description
v	DMARC1	Version	Identifies the record retrieved as a DMARC record. It must be the first tag in the list.
p	quarantine	Policy	Policy to apply to email that fails the DMARC test. Valid values can be 'none', 'quarantine', or 'reject'.
fo	1	Forensic Reporting	Provides requested options for generation of failure reports. Valid values are any combination of characters '01ds' seperated by ':

Test	Result
 DMARC Record Published	DMARC Record found
 DMARC Syntax Check	The record is valid
 DMARC Multiple Records	Multiple DMARC records corrected to a single record.
 DMARC Policy Not Enabled	DMARC Quarantine/Reject policy enabled
 DMARC External Validation	All external domains in your DMARC record are giving permission to send them DMARC reports.

`v=DMARC1` indicates the version of DMARC required and is also a mandatory tag. `p=quarantine` indicates the DMARC policy is set to quarantine, meaning that any email failing the DMARC test will be sent to spam as per the set policy. `fo=1` enables forensic reporting, generating a report on any failures.

All of the previous steps were done just to get information on the email. Now it was time to investigate the actual attachment itself. I wanted to investigate the file and determine if it had malware by using its hash vaule. So, I downloaded the attachment into the virtual environment and ran the `sha256sum` command via the terminal.

```
ubuntu@tryhackme:~$ cd Desktop
ubuntu@tryhackme:~/Desktop$ ls
SWT_#09674321____PDF____.CAB  challenge.eml  mate-terminal.desktop
ubuntu@tryhackme:~/Desktop$ sha256sum ^C
ubuntu@tryhackme:~/Desktop$ sha256sum SWT_#09674321____PDF____.CAB
2e91c533615a9bb8929ac4bb76707b2444597ce063d84a4b33525e25074fff3f  SWT_#09674321_
____PDF____.CAB
ubuntu@tryhackme:~/Desktop$ █
```

I then opened up VirusTotal on my main host machine and used the attachment's hash value. VirusTotal ended up flagging 48 of the 63 vendors tied to the attachment, marking it as a malicious file, labelling the file as trojan, ransomware, and spyware.

48

/ 63

Community Score

1

48/63 security vendors flagged this file as malicious

2e91c533615a9bb8929ac4bb76707b2444597ce063d84a4b33525e25074fff3f

SWT_#09674321___.PDF___.CAB

rar attachment spreader

DETECTION

DETAILS

RELATIONS

COMMUNITY 5

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label trojan.msil/loki

Threat categories trojan ransomware spyware

Security vendors' analysis

AhnLab-V3	Trojan/Win32.Kryptik.R345359	AliCloud
ALYac	Gen:Variant.Ransom.Loki.8140	Antiy-AVL
Arcabit	Trojan.Ransom.Loki.D1FCC	Avast
AVG	Win32:MalwareX-gen [Pws]	Avira (no cloud)
BitDefender	Gen:Variant.Ransom.Loki.8140	Bkav Pro
ClamAV	Legacy.Trojan.Agent-1388784	CTX
Cynet	Malicious (score: 99)	DeepInstinct
DrWeb	Trojan.PackedNET.331	Emsisoft
eScan	Gen:Variant.Ransom.Loki.8140	ESET-NOD32

Some final details that VirusTotal gave me were the size and extension of the attached file. The file size was 400.26 KB, and the file was really a .rar type file.

Classification:

As identified by VirusTotal, the attached .rar file has trojans, ransomware, and spyware, making this email a True Positive for phishing.

Key Takeaways:

- When observing the source of an email, in most modern systems, the order of the message headers from top to bottom is newest to oldest. However, some older systems may show it in reverse.
- SPF, DKIM, and DMARC are important indicators letting one know the authenticity of emails.
- To do an SPF or DMARC lookup, you need the Return-Path domain of the email that you are investigating, usually found in the email's source